

Jae Ho Choi

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PREFERRED NAME	“Jay” or “Jae”
EMPLOYMENT	Department of Mathematical Sciences, Carnegie Mellon University, Pittsburgh, Pennsylvania, USA Postdoctoral Associate Sep 2024 – Present
EDUCATION	University of Pennsylvania, Philadelphia, Pennsylvania, USA - Ph.D. in Mathematics Sep 2019 – May 2024 - Advisors: Robert M. Strain, Yoichiro Mori - Master of Computer and Information Technology (MCIT) Sep 2021 – May 2023 University of Maine, Orono, Maine, USA - M.A. in Mathematics Sep 2017 – May 2019 - Thesis: Theory of Lexicographic Differentiation in the Banach Space Setting - Advisor: Peter Stechliniski Williams College, Williamstown, Massachusetts, USA - B.A. in Physics, Mathematics, and Economics Sep 2013 – May 2017 - Senior Thesis (in Physics): An Unusual Derivation of the Boltzmann Distribution for a Particle in a 1D Box - Advisor: William K. Wootters - Graduated with Departmental Honors in Physics
CITIZENSHIP	South Korean, with U.S. permanent residency
PERSONAL WEBSITE	choijh93.github.io
RESEARCH INTERESTS	partial differential equations arising in fluid mechanics, with primary focus on well-posedness, stability of solutions, and long-term dynamics of free boundary problems, and secondary focus on numerical simulation of free boundary problems
PREPRINTS	[9] J. Choi and I. Tice, Traveling wave solutions to a two-phase Stokes problem with viscosity contrast driven by surface tension in 2D, <i>in preparation</i>. [8] J. Choi and R. Strain, Stability of a two-phase Taylor-Melcher problem with viscosity contrast driven by surface tension in 2D, <i>in preparation</i>. [7] J. Choi, Critical global well-posedness for the two-phase Brinkman problem with surface tension, <i>submitted</i>. [6] L. Bouck, J. Choi, and N. Walkington, A stable boundary integral scheme for a two-phase Brinkman problem in 2D, <i>in preparation</i>. [5] J. Choi and R. Strain, Stability of a two-phase Stokes problem with viscosity contrast driven by surface tension in 2D, <i>in preparation</i>. [4] J. Choi and I. Tice, Traveling wave solutions to a general incompressible Navier-Stokes-Fourier system with free boundary, <i>submitted</i>. [3] J. Choi, Stability of a two-phase Stokes problem with surface tension, <i>submitted</i>.
PUBLICATIONS	[2] J. Choi, N. Krishna, N. Magill, and A. Sarria, On the L^p regularity of solutions to the generalized Hunter-Saxton system, <i>Discrete Contin. Dyn. Syst. Ser. B</i> 24 (2019), 6349-6365. [1] S. Alterman, J. Choi, R. Durst, S. Fleming, and W. Wootters, The Boltzmann distribution and the quantum-classical correspondence, <i>J. Phys. A</i> 51 (2018), no. 34.
INVITED TALKS	Traveling wave solutions to a general incompressible Navier-Stokes-Fourier system with free boundary. Korea Institute for Advanced Study, Seoul, Republic of Korea Aug 2026

	▪ <i>A two-phase Stokes problem with viscosity contrast.</i> Graduate Student and Postdoc (GSPS) Seminar Department of Mathematical Sciences, Carnegie Mellon University, PA, USA Oct 2025 ▪ <i>On the two-phase Stokes flow problem with surface tension.</i> Recent Progress in PDE Models of Incompressible Fluids AMS Western Sectional Meeting, University of Denver, CO, USA Aug 2025 ▪ <i>Stability and instantaneous analyticity of a 2D Stokes bubble with viscosity contrast.</i> PDE and Applied Analysis Seminar Pohang University of Science and Technology, Pohang, Republic of Korea Jul 2025 ▪ <i>Well-posedness of a two-phase Stokes flow problem with surface tension.</i> Center for Nonlinear Analysis (CNA) Seminar Carnegie Mellon University, PA, USA Sep 2024 ▪ <i>On the two-phase Stokes flow problem with surface tension.</i> PDE and Applied Analysis Seminar Pohang University of Science and Technology, Pohang, Republic of Korea Jun 2024 ▪ <i>The mathematics of fluids.</i> UP GRADe (an NSF-funded talk series co-hosted by Penn's AWM chapter and GeMs in Math) Department of Mathematics, University of Pennsylvania, PA, USA May 2024 ▪ <i>The boundary integral method for interfacial Stokes flow.</i> Analysis Seminar, Department of Mathematics, University of Pennsylvania, PA, USA Apr 2024 ▪ <i>On the two-phase Stokes flow problem with surface tension.</i> Analysis Seminar, Department of Mathematics, University of Pennsylvania, PA, USA Nov 2023 ▪ <i>On the two-phase Stokes flow problem with surface tension.</i> Applied and Numerical PDE Seminar, University of California, Berkeley, CA, USA Nov 2023 ▪ <i>A friendly introduction to the Taylor-Melcher model.</i> Recent Advances in Mathematical Fluid Dynamics, Duke University, NC, USA May 2023 ▪ <i>Infinite-energy solutions of the Euler equations with time-dependent damping.</i> Williams College REU Conference, Williams College, MA, USA Jul 2016 ▪ <i>Reduction of error in the discrete truncated Wigner approximation.</i> Williams College Summer Science Research Poster Session, Williams College, MA, USA Aug 2015
INVITED CONFERENCES/ WORKSHOPS & RESEARCH VISITS	▪ POSTECH-KIAS Intensive Lecture Series on PDEs I & II, Republic of Korea Jun 2026 ▪ Conference on Kinetic Theory and Related Fields, Republic of Korea Jun 2026 ▪ Winter School: Boundary and Singularity in Fluid Mechanics, Simons Center for Geometry and Physics, NY, USA Jan 2025 ▪ Basque Center for Applied Mathematics (BCAM), Bilbao, Spain Dec 2024 ▪ Mathematics of General Relativity and Fluids (SLMath), Heraklion, Crete, Greece Jul 2024 ▪ Summer School on Wave Turbulence, Massachusetts Institute of Technology, MA, USA Jul 2023 ▪ New Trends in Mathematical Fluid Dynamics, Institut Fourier, Gières, France Jun 2023 ▪ Recent Advances in Mathematical Fluid Dynamics, Duke University, NC, USA May 2023
SERVICE & STUDENT MENTORSHIP	▪ Panelist, Graduate School Panel (hosted by the Student Mathematics and Statistics Advisory Board) Department of Mathematics, Williams College, MA, USA Oct 2024 ▪ Mentor, Directed Reading Program (DRP), Department of Mathematics, University of Pennsylvania The DRP is an initiative that pairs undergraduates with graduate or postdoctoral mentors for semester-long independent studies. • Darren Zheng, harmonic analysis (Spring 2024). • Dylan Marchlinski, fluid dynamics and probability theory (Fall 2023). • Evan Qiang, probability theory (Spring 2023). • David Kogan, probability theory (Spring 2022).
TEACHING EXPERIENCE	CARNEGIE MELLON UNIVERSITY ▪ Instructor, Integration and Approximation (21-122) Fall 2024 ▪ Instructor, Integration and Approximation (21-122) Spring 2025 ▪ Instructor, Integration and Approximation (21-122) Fall 2025 ▪ Co-Instructor, Integration and Approximation (21-122) Spring 2026

- Instructor, Principles of Real Analysis II (21-356) Fall 2026
- UNIVERSITY OF PENNSYLVANIA**
- Teaching Assistant, Advanced Analysis (MATH 5090) Spring 2024
 - Teaching Assistant, Advanced Analysis (MATH 5080) Fall 2023
 - Grader, Mathematics in the Age of Information (MATH 2100) Spring 2022
 - Teaching Assistant, Ideas in Mathematics (MATH 1700) Spring 2021
 - Teaching Assistant, Advanced Calculus (MATH 3610) Fall 2020

UNIVERSITY OF MAINE

- Teaching Assistant, Calculus II (MAT 127) Spring 2019
- Teaching Assistant, Calculus I (MAT 126) Fall 2018
- Teaching Assistant, Calculus II (MAT 127) Spring 2018
- Teaching Assistant, Calculus II (MAT 127) Fall 2017

WILLIAMS COLLEGE

- Teaching Assistant, Real Analysis (MATH 350) Spring 2016
- Teaching Assistant, Linear Algebra (MATH 250) Fall 2015

**AWARDS &
SCHOLARSHIPS**

- **Good Teaching Award**
Department of Mathematics, University of Pennsylvania Fall 2023, Spring 2024
- **Benjamin Franklin Fellowship**
Department of Mathematics, University of Pennsylvania Sep 2019 – May 2024
- **Class of 1960 Scholar in Physics**
Department of Physics, Williams College Sep 2016 – May 2017
- **John & Louise Finnerty Class of 1971 Fund for Applied Mathematical Research**
Department of Mathematics and Statistics, Williams College Jun 2016– Aug 2016